Introduction to Nonparametric Statistics
Problem Set 4: Density Estimation by Kernel Estimators

Exercise 1: Let f be a density belonging to the Hölder class $\Sigma(\beta, L)$ and let $(X_1, \dots, X_n)$ be a sample of size n of random variables with density f. For every integer $s < \lfloor \beta \rfloor$, define the estimator of $f^{(s)}$, the derivative of order s of f, by setting, for $h > 0$,

$$\hat{f}_{n,s}(x) = \frac{1}{nh^{s+1}} \sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right), \quad x \in \mathbb{R}$$

where K is a function such that for every $j \in \{0, 1, \dots, \lfloor \beta \rfloor\}$ the function $u \mapsto u^j K(u)$ is integrable and for every $j \in \{0, \dots, \lfloor \beta \rfloor\} \setminus \{s\}$,

$$\|K\|_2 < \infty, \quad \int u^j K(u) du = 0 \quad \text{and} \quad \int u^s K(u) du = s! \tag{1}$$

$$\int |u|^\beta K(u)|du < \infty$$

1. Show that for every $x_0 \in \mathbb{R}$, there exist two constants C_B and C_V such that for every $h > 0$,
$$B(\hat{f}_{n,s}(x_0)) := |\mathbb{E}[\hat{f}_{n,s}(x_0)] - f^{(s)}(x_0)| \le C_B h^{\beta - s},$$
$$\text{var}(\hat{f}_{n,s}(x_0)) := \mathbb{E}\left[\left(\hat{f}_{n,s}(x_0) - \mathbb{E}[\hat{f}_{n,s}(x_0)]\right)^2\right] \le \frac{C_V}{nh^{2s+1}}.$$
2. Show that there exists a constant $C > 0$ such that, if h is chosen correctly,
$$\mathbb{E}\left[\left(\hat{f}_{n,s}(x_0) - f^{(s)}(x_0)\right)^2\right] \le C n^{-\frac{2(\beta - s)}{2\beta + 1}}.$$

Exercise 2: We observe a sample of size n, $(X_1, Y_1), \dots, (X_n, Y_n)$, having the same distribution as a pair of real-valued random variables (X, Y). We assume that the vector (X, Y) has density f such that for every $x \in \mathbb{R}$ and every $y \in \mathbb{R}$, $f(x, y) > 0$. For any two functions f_1 and f_2 of two variables, we denote by $\star$ the convolution product between f_1 and f_2 whenever it is defined: for every $(x, y) \in \mathbb{R}^2$,

$$(f_1 \star f_2)(x, y) = \iint f_1(x - u, y - v) f_2(u, v) du dv.$$

We consider $\hat{f}_h$, the estimator of f defined for every $(x, y) \in \mathbb{R}^2$ by:

$$\hat{f}_h(x, y) = \frac{1}{nh_1h_2} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h_1}, \frac{y - Y_i}{h_2}\right),$$

where $h = (h_1, h_2)$ with $h_1 > 0$, $h_2 > 0$, and $K : \mathbb{R}^2 \mapsto \mathbb{R}$ is a kernel (hence $\iint K(u, v) du dv = 1$) such that $\iint |K(u, v)||v| du dv < \infty$, $\iint |K(u, v)||u|^{1/2} du dv < \infty$, and such that $\|K\|_2^2 = \iint K^2(u, v) du dv < \infty$. We then set, for every $(x, y) \in \mathbb{R}^2$,

$$K_h(x, y) = \frac{1}{h_1h_2} K\left(\frac{x}{h_1}, \frac{y}{h_2}\right).$$

1. Show that the pointwise quadratic risk at $(x, y) \in \mathbb{R}^2$ of $\hat{f}_h$ can be written as:

$$\mathbb{E}\left[(\hat{f}_h(x,y) - f(x,y))^2\right] = \left(\mathbb{E}[\hat{f}_h(x,y)] - f(x,y)\right)^2 + \mathrm{var}(\hat{f}_h(x,y)).$$

2. Show that for $x \in \mathbb{R}$ and $y \in \mathbb{R}$, $\mathbb{E}[\hat{f}_h(x,y)] = (K_h \star f)(x,y)$.

3. Compute, for $x \in \mathbb{R}$ and $y \in \mathbb{R}$, $\mathrm{var}(\hat{f}_h(x,y))$ as a function of n, $K_h \star f$, and $K_h^2 \star f$.

4. Show that if f is bounded by M, then

$$\mathrm{var}(\hat{f}_h(x,y)) \leq \frac{M\|K\|_2^2}{nh_1h_2}.$$

5. Show that if f also satisfies the following property: for all $(u,v) \in \mathbb{R}^2$ and $(u',v') \in \mathbb{R}^2$,

$$\left|f(u,v) - f(u',v')\right| \leq |u-u'|^{1/2} + |v-v'| \tag{2}$$

then $B_h(x,y)$, the bias of $\hat{f}_h(x,y)$, satisfies for $x \in \mathbb{R}$ and $y \in \mathbb{R}$:

$$B_h(x,y) \leq C_K\left(h_1^{1/2} + h_2\right),$$

where C_K is a constant depending only on K.

6. We still assume that f is bounded and satisfies (2). Determine the optimal values of h_1 and h_2, up to constants, in order to obtain the rate of the estimator $\hat{f}_h$ for the pointwise quadratic risk. Give the value of this rate, up to constants.

Exercise 3

Notation

- if μ and ν are measures, the notation $\mu \ll \nu$ means that μ is absolutely continuous with respect to ν. Recall that this means: $\forall A \in \mathcal{F}$, if $\nu(A) = 0$ then $\mu(A) = 0$.

- for a real number x, we denote by $x_- = \max(-x, 0)$ its negative part and by $x_+ = \max(x, 0)$ its positive part. Of course, $x = x_+ - x_-$ and $|x| = x_+ + x_-$.

Kullback–Leibler divergence

Let P and Q be two probability measures defined on a measurable space $(\Omega, \mathcal{F})$, and let μ be a common reference measure such that $P \ll \mu$ and $Q \ll \mu$. We denote by $p = \frac{dP}{d\mu}$ and $q = \frac{dQ}{d\mu}$ the densities of P and Q with respect to μ.

The Kullback–Leibler divergence (or *KL divergence*) of P with respect to Q is defined by:

$$D_{\mathrm{KL}}(P\|Q) = \begin{cases} \int_\Omega \log\left(\frac{p(x)}{q(x)}\right) p(x)\, d\mu(x) & \text{if } P \ll Q \\ +\infty & \text{otherwise} \end{cases}$$

It is a measure of the "gap" between the two distributions.

<u>Remarks</u>

- This definition can also be written in the form

$$D_{\mathrm{KL}}(P\|Q) = \int_{\Omega} \log\left(\frac{dP}{dQ}\right)\, dP,$$

when $P \ll Q$, where $\frac{dP}{dQ}$ is the Radon–Nikodym density of P with respect to Q.

- When $P \ll Q$, we have, μ-almost everywhere: $p(x) > 0 \Rightarrow q(x) > 0$ (which guarantees that the ratio p/q is well defined whenever $p > 0$).

- KL divergence is not a distance in the metric sense: it is not symmetric and does not satisfy the triangle inequality.

- **We are going to show that this definition always makes sense and that one always has $D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality if and only if $P = Q$.**

- Special case: $\Omega = \{1, 2, \ldots, d\}$. The probabilities can then be identified with vectors in $\mathbb{R}^d$: $p = (p_1, \ldots, p_d)$ and $q = (q_1, \ldots, q_d)$. If, for every i, $q_i = 0$ implies $p_i = 0$[1], then we have

$$D_{\mathrm{KL}}(p\|q) = \sum_{i=1}^{d} p_i \log\left(\frac{p_i}{q_i}\right) = H(p,q) - H(p)$$

where $H(p,q) = -\sum_{i=1}^{d} p_i \log(q_i)$ is the cross-entropy[2] and $H(p) = -\sum_{i=1}^{d} p_i \log p_i$ is the entropy of p.

Part 1

For simplicity (and to avoid certain technical complications), we assume that q and p are strictly positive μ-almost surely. Then $P \ll Q$ and $Q \ll P$.

The result of Question 1 remains true if one only assumes that $P \ll Q$, and the results of Questions 2 and 3 are true without any assumption.

1. Show that $\int \left(p \log(\frac{p}{q})\right)_{-} d\mu$ is finite.

   Remark: this shows that the Kullback–Leibler divergence is always well defined. Indeed, if Y is a random variable such that $E(Y_-) < \infty$, then:

   - Either $E(Y_+) < \infty$, in which case $E(|Y|) < \infty$, i.e. Y is integrable.
   - Or $E(Y_+) = +\infty$, in which case $E(|Y|) = +\infty$, but one can still make sense of $E(Y)$ by setting $E(Y) = +\infty$.

2. Show that $D_{\mathrm{KL}}(P\|Q) \geq 0$ for every P and every Q. (must know)

3. Show that $D_{\mathrm{KL}}(P\|Q) = 0$ is equivalent to $P = Q$. (must know)

Part 2

Let $X_1, \ldots, X_n$ be an i.i.d. sample with distribution P. We assume that P is absolutely continuous with respect to μ with unknown density f:

$$P = f d\mu.$$

[1] this is exactly the assumption of absolute continuity of p with respect to q

[2] "cross-entropy", see the statistical learning course

We model this density by q_θ with $\theta \in \Theta$ (for some set $\Theta \in \mathbb{R}^d$). In other words, we assume that

$$\text{there exists } \theta_0 \text{ such that } q_{\theta_0} \text{ is "close" to } f$$

We must clarify the meaning of "close".

We choose the Kullback–Leibler divergence as our measure of proximity. We therefore seek θ_0 such that

$$\theta_0 \in \arg\min_{\theta \in \Theta} D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$$

Problem: we do not know f, so we do not have access to $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$ either. We therefore seek to replace this unknown quantity $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$ by a quantity that is close to it (in some sense) but computable from the data alone. To do this, we observe that the following quantity

$$\hat{D}(\theta) := -\frac{1}{n} \sum_{i=1}^{n} \log \frac{q_\theta(X_i)}{f(X_i)}$$

is close (in some sense) to $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$ (this is the point of Questions 2 and 3). This quantity still depends on f. But one immediately notices that

$$\arg\min_{\theta \in \Theta} -\frac{1}{n} \sum_{i=1}^{n} \log \frac{q_\theta(X_i)}{f(X_i)} = \arg\min_{\theta \in \Theta} -\frac{1}{n} \sum_{i=1}^{n} \log q_\theta(X_i)$$

We therefore decide to estimate θ_0 by

$$\hat{\theta} = \arg\min_{\theta \in \Theta} -\frac{1}{n} \sum_{i=1}^{n} \log q_\theta(X_i)$$

1. The estimator $\hat{\theta}$ is a very well-known estimator: which one is it?
2. Assume that $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$ is finite. Show that $\hat{D}(\theta)$ is an unbiased and consistent estimator of $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$, i.e. show that
$$E(\hat{D}(\theta)) = D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$$
and that
$$\hat{D}(\theta) \xrightarrow[n \to +\infty]{p.s.} D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$$
3. (optional question) Assume that $f d\mu \ll q_\theta d\mu$ but that $D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu) = +\infty$.
   (a) Let $Y_1, \ldots, Y_n$ be positive i.i.d. random variables such that $E(Y_1) = +\infty$. Show that $\bar{Y}_n \xrightarrow[n \to +\infty]{p.s.} +\infty$.
   (b) Show that one still has
$$\hat{D}(\theta) \xrightarrow[n \to +\infty]{p.s.} D_{\mathrm{KL}}(f d\mu \| q_\theta d\mu)$$

Part 3

So far, we have worked in a parametric setting. But the ideas and the proofs would have been the same in a more general setting.

For instance, to keep things as simple as possible, suppose that we are hesitating between several possible densities $f_1, \ldots, f_p$ in order to estimate the unknown density f. In other words, let us place ourselves in an idealized situation where we know that the unknown density f is equal to one of these p densities (or is close to one of these p densities), these p densities being arbitrary, i.e. not necessarily coming from a particular family of distributions.

We can then choose the estimator $f_{\hat{j}}$ with $\hat{j}$ satisfying

$$\hat{j} \in \arg\min_{j\in\{1,\dots,p\}} -\frac{1}{n}\sum_{i=1}^{n}\log\frac{f_j(X_i)}{f(X_i)} = \arg\max_{j\in\{1,\dots,p\}}\sum_{i=1}^{n}\log f_j(X_i)$$

In the end: there is no need to have a parametric model in order to think about maximizing the log-likelihood in the context of estimating an unknown density. The reference densities can be arbitrary.

One might then think of using log-likelihood maximization to choose the bandwidth h of a KDE. The following questions concern this point.

Suppose we have at our disposal a collection of KDEs $(\hat{f}_h)_{h\in H}$ where H is a grid of possible values

These KDEs have been fitted using an i.i.d. sample with distribution P and density $fd\lambda$, where λ is Lebesgue measure on $\mathbb{R}$ and f is the unknown density that we wish to estimate. We assume that K is positive.

We want to find the "best" possible bandwidth in the grid. To do this, we need to choose a measure of proximity. We choose the Kullback–Leibler divergence: ideally, we would like to find a bandwidth h_0 such that

$$h_0 \in \arg\min_{h>0} D_{\mathrm{KL}}(fd\lambda \| \hat{f}_h d\lambda)$$

As before, this "ideal" bandwidth h_0 minimizes an unknown quantity because it depends on the unknown density f. We would like to compute, from the data alone, a bandwidth $\hat{h}$.

Technical assumption: we assume that (a.s.) $fd\lambda \ll \hat{f}_h d\lambda$.

We could use the ideas of the previous paragraph (choice of the index j) and choose the bandwidth $\hat{h}_{\text{bad-idea}}$ such that

$$\hat{h}_{\text{bad-idea}} \in \arg\max_{h>0}\sum_{i=1}^{n}\log\hat{f}_h(X_i)$$

1. Explain why this is not a good idea.

2. We decide to split our sample $X_1,\dots,X_n$ into two parts, forming a so-called training sample (training set) and a validation sample (validation set). Let I denote the indices corresponding to the training sample and J the indices corresponding to the validation sample.

   For each value in the grid H, we build a KDE using the training sample. We therefore have at our disposal a collection of estimators $(\hat{f}_h)_{h\in H}$, and these estimators have thus been computed from $(X_i)_{i\in I}$. We decide to choose the bandwidth $\hat{h}$ such that

   $$\hat{h} \in \arg\max_{h>0}\sum_{i\in J}\log\hat{f}_h(X_i)$$

   (note that the sum is therefore taken over the indices corresponding to the validation sample!!)

   Compute

   $$E\Big(\frac{-1}{Card(J)}\sum_{i\in J}\log\frac{\hat{f}_h(X_i)}{f(X_i)} \mid \{X_i\}_{i\in I}\Big)$$

   (You should then understand the value of using a validation sample to choose the parameter h...)

   Then we use the same ideas as in the learning course for cross-validation.